Questions

Q1.

A man throws a stone into a pond.

On the other side of the pond, the water becomes very shallow.

In the shallow water, the wave is slower but the frequency does not change.

State what happens to the wavelength when a wave reaches the shallow water.

(1)

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.....

(Total for question = 1 mark)

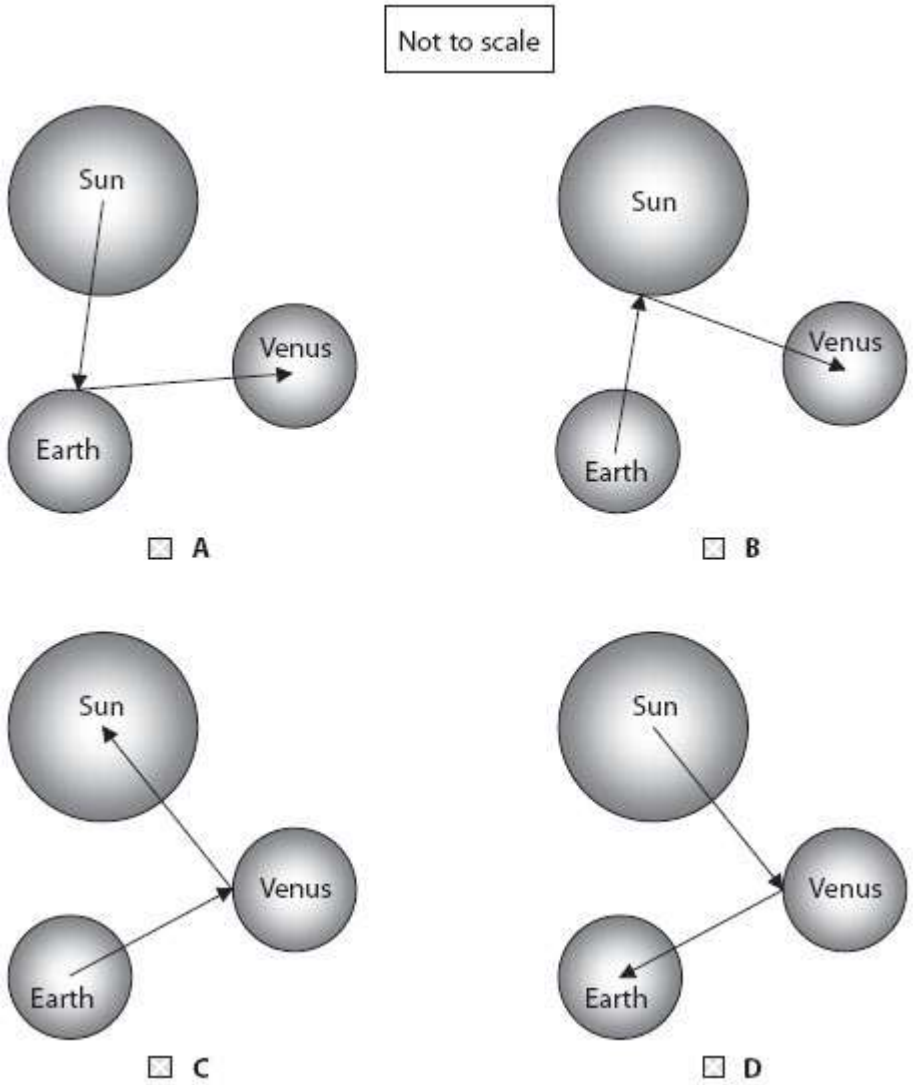
Q2.

(a) Galileo was one of the first scientists to use a telescope to study Venus.

(i) Which of these diagrams best shows how light waves enable us to see Venus?

Put a cross ( ) in the box next to your answer.

(1)



(ii) Use words from the box to complete the sentences.

(2)

asteroids	comets	geocentric	heliocentric
	moons	particle	stars

Galileo also used his telescope to observe the ..... of Jupiter.
His observations provided evidence to support the ..... model of the Solar System.

(iii) Describe how a reflecting telescope is different from the simple telescope which Galileo used.

(2)

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(b) (i) The diagram represents a wave.

[redacted]

State the amplitude and wavelength of the wave.

(2)

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(ii) 20 waves are sent out in 4 seconds.

Complete the sentence by putting a cross ( ) in the box next to your answer.

The frequency of the wave is

(1)

☐ **A** 0.2 Hz

☐ **B** 5 Hz

☐ **C** 20 Hz

☐ **D** 80 Hz

(Total for Question is 8 marks)

Q3.

(i) Figure 13 shows a student sitting on the shore of a lake watching ripples on the surface of the water moving past a toy boat.

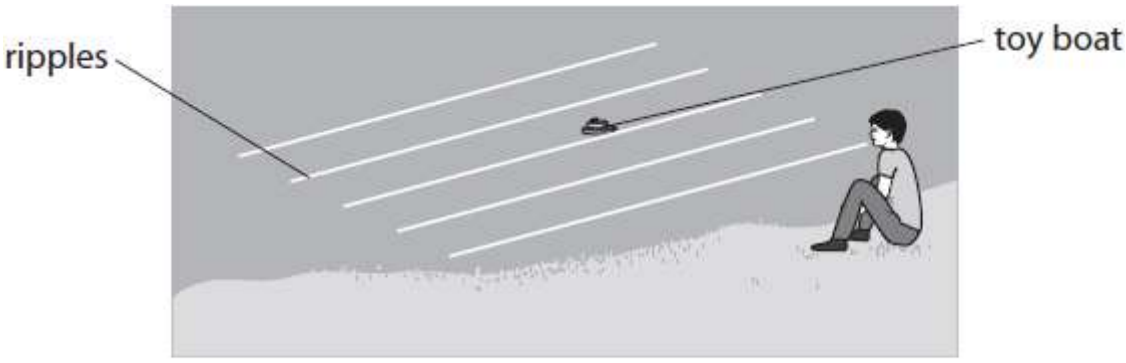

Figure 13

The student has a stopwatch.
Describe how the student could determine the frequency of the ripples on the lake.

(3)

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(ii) The speed of a water wave is 1.5 m/s.

The frequency of the wave is 0.70 Hz.
Calculate the wavelength of this wave.
Use the equation

$$v = f \times \lambda$$

(2)

wavelength = ... m

(iii) Water waves are transverse waves.

Describe the difference between transverse waves and longitudinal waves.

(2)

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(Total for question = 7 marks)

Q4.

Figure 9 is a diagram of a water wave.

A cork is floating on the water.

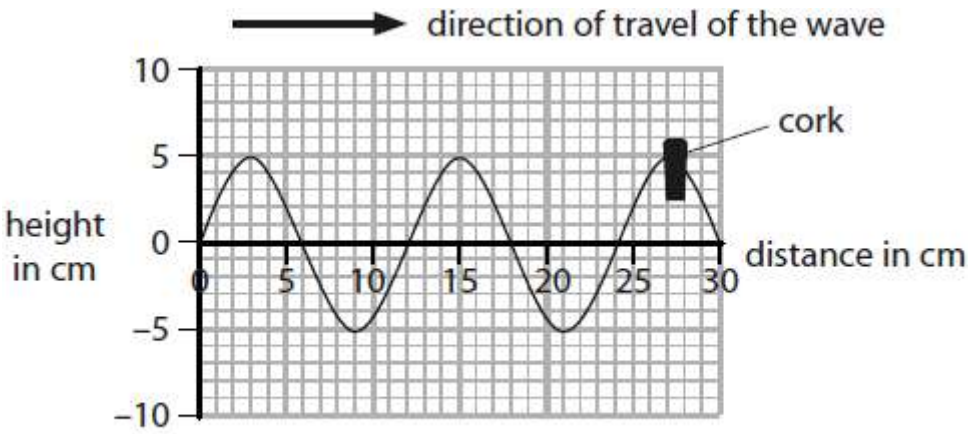

Figure 9

(i) Use the scale on the diagram to measure the wavelength of the wave.

(2)

wavelength = ... cm

(ii) Describe the motion of the cork.

You should include how the cork moves relative to the direction of travel of the wave.

(2)

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(Total for question = 4 marks)

Q5.

Figure 4 shows a wave on the sea moving towards a beach.

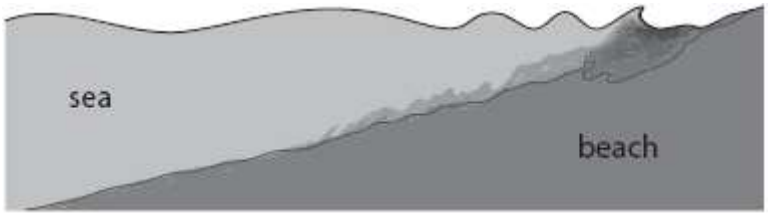

Figure 4

(i) State the relationship between the speed, frequency and wavelength of a wave.

(1)

.....

.....

.....

.....

(ii) A water wave has a frequency of 0.40 Hz and a wavelength of 1.2 m.

Calculate the speed of the water wave.

(2)

speed = ... m / s

(Total for question = 3 marks)

Q6.

This question is about waves.

Figure 3 is a diagram of a **water wave** in a ripple tank.

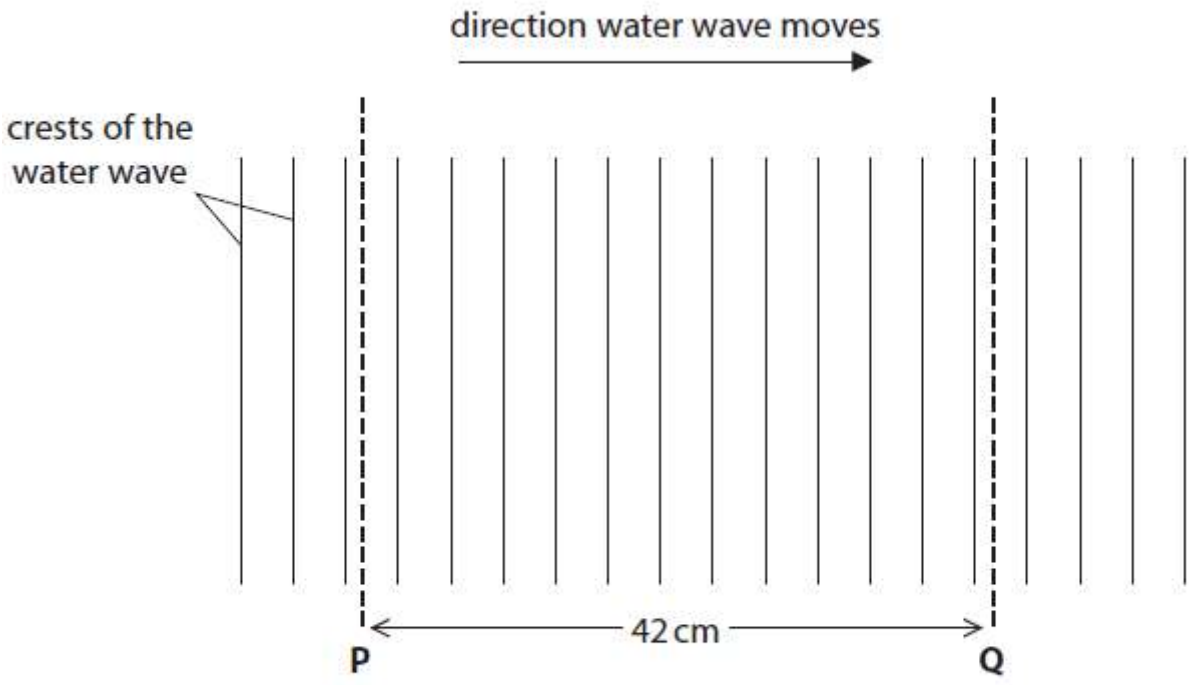

Figure 3

(i) State the number of crests of the wave between **P** and **Q**.

(1)

number of crests= .....

(ii) The distance between **P** and **Q** is 42 cm.

Calculate the wavelength of the water wave in Figure 3.

(2)

wavelength= ..... cm

(iii) Describe how a student could determine the wave speed of the water wave in Figure 3.

(3)

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(Total for question = 6 marks)

Q7.

When an earthquake happens, waves are transmitted through the Earth.
Figure 17 is a diagram of a part of the Earth's crust.

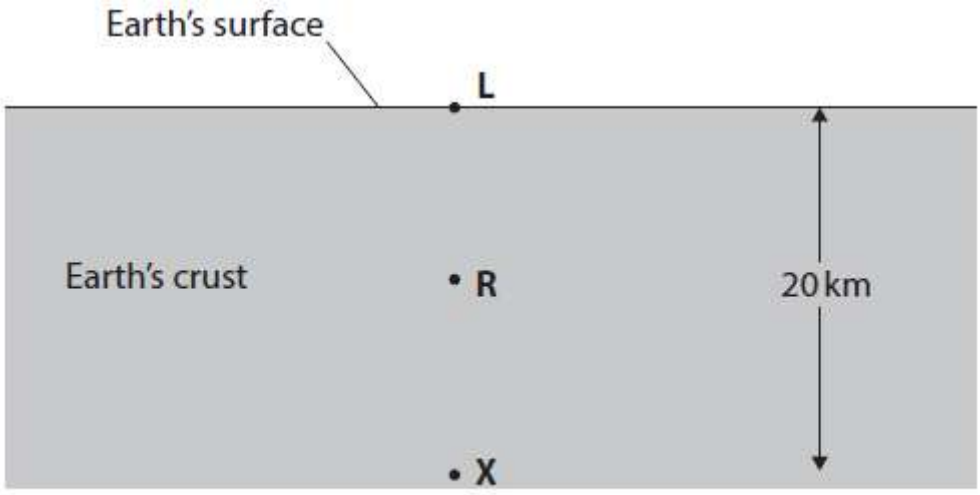

Figure 17

An earthquake happens at point **X**, 20 km below the Earth's surface.

An earthquake wave from **X** is detected at point **L**.

This earthquake wave is a transverse wave.

(i) State **one** other example of a transverse wave.

(1)

.....

(ii) A point in the crust is labelled **R** in Figure 17.

There is a rock at point **R**.
Describe the effect that this earthquake wave has on the rock at **R** as the wave passes through the rock.
You may add to Figure 17 to help your answer.

(2)

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.....

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.....

(Total for question = 3 marks)

Q8.

A sound wave in air travels a distance of 220 m in a time of 0.70 s.

(i) State the equation linking speed, distance and time.

(1)

(ii) Calculate the speed of the sound wave in air.

(2)

wave speed = ... m/s

(Total for question = 3 marks)

Q9.

Figure 13 shows part of the inside of the Earth below the surface.

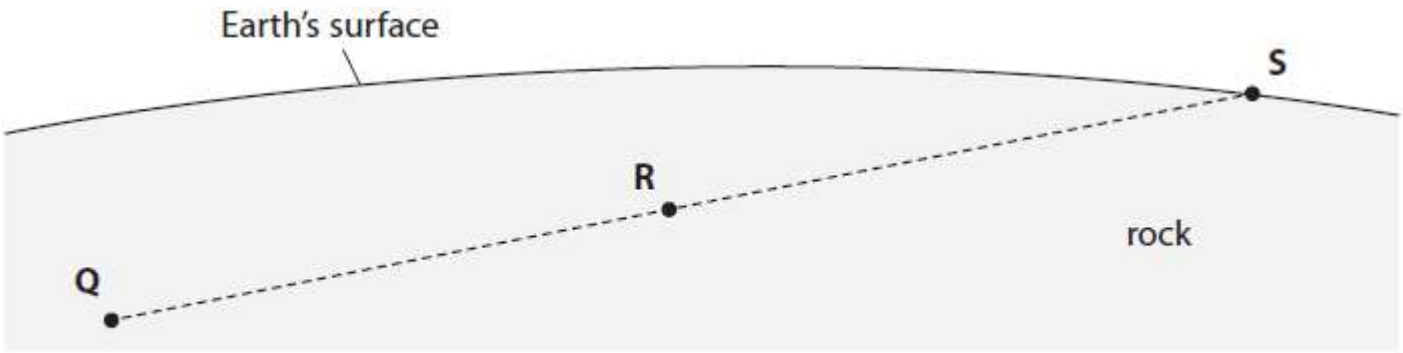


Figure 13

An earthquake starts at **Q**.

A seismic wave travels from **Q** to **S**.

The seismic wave is a longitudinal wave.

(i) Draw arrows on Figure 13 to show how the rock at **R** moves when the seismic wave passes through **R**.

(2)

(ii) The frequency of the seismic wave is 12 Hz.

The wave speed of the seismic wave is 7 km / s.
Calculate the wavelength of the seismic wave, in metres.
Use the equation

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

(3)

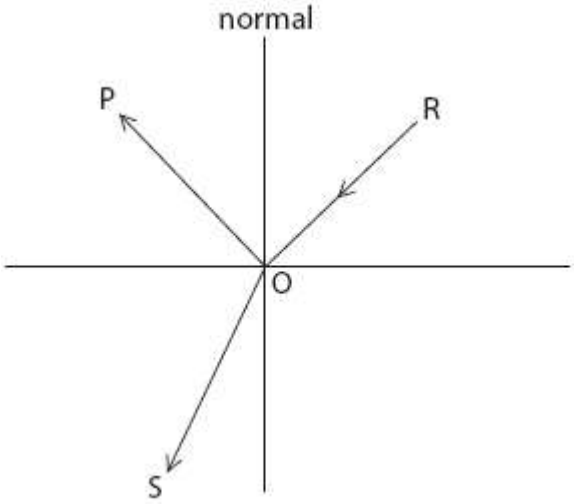
wavelength = ... m

(Total for question = 5 marks)

Q10.

When light strikes a glass surface it can be both refracted and reflected.

(a) The diagram shows the possible paths for a ray of light which strikes a surface at the point O.



(i) Which of the lines show the possible path of a ray of light passing from air into glass?
Put a cross ( ) in the box next to your answer.

(1)

☒ **A** POS

☒ **B** POR

☒ **C** ROP

☒ **D** ROS

(ii) The diagram shows a water wave going from deep water into an area of much shallower water. The wave is refracted at the boundary between deep water and shallow water.

Which row of the table is correct for what happens when the wave is refracted?
Put a cross ( ) in the box next to your answer.

(1)

	speed	direction
☐ **A**	stays the same	changes
☐ **B**	stays the same	stays the same
☐ **C**	changes	changes
☐ **D**	changes	stays the same

(b) In 1610 Galileo used a refracting telescope to observe the planet Jupiter..
(i) Explain how a refracting telescope produces a magnified image of Jupiter.

(3)

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(ii) In 1610, the geocentric model of the Solar System was commonly accepted.

Explain how Galileo's observations contradicted the geocentric model.

(3)

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(c) Light travels the 150 million km from the Sun to the Earth in about 500 s.
It takes about 2100 s for light to reach the Earth from Jupiter.
Using this information, calculate the approximate distance of Jupiter from the Earth.

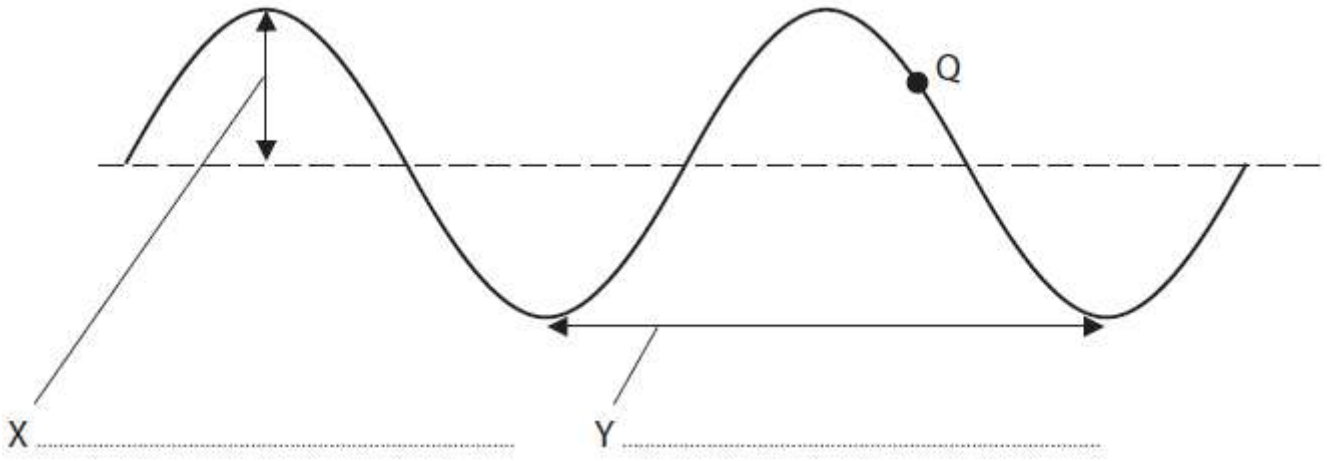
(2)

distance of Jupiter from the Earth =.....million km

(Total for Question = 10 marks)

Q11.

The diagram shows a transverse wave.



(i) Use words from the box to label the distances X and Y.

amplitude	frequency	magnification	speed	wavelength

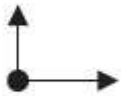
(2)

(ii) Q is a particle in the wave.

Which of these shows the way in which particle Q moves?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A**

☐ **B** 

☐ **C**

☐ **D**

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
	wavelength gets shorter / decreases		(1)

Q2.

	Answer	Acceptable answers	Mark
(a)(i)	D		**(1)**
(a)(ii)	moons (1) heliocentric (1)	must be in correct order	**(2)**
(a)(iii)	A description including two of the following points Reflecting telescope has mirror(s) (1) Galilean telescope has only lenses (1) Reflecting telescope can gather more light / can have a larger objective (1) Image viewed from the side of reflecting telescope (1) Image viewed from end of Galilean telescope. (1)	refracting telescope reverse argument	**(2)**
(b)(i)	5 (cm) (1) 8 (cm) (1)	+5 -5 0.08 m 80 mm	**(2)**
(b)(ii)	B		**(1)**

Q3.

Question number	Answer	Additional guidance	Mark
(i)	a description to include count the number of waves(1) (arriving/passing a point) in a specific time(1) use frequency = $\frac{\text{number of waves}}{\text{time}}$ (1)	 ignore in one second count the number of waves in one second scores 2 marks (MP1 and MP3) find the time between one wave and the next scores 2 marks (MP1 and MP2)	**(3)** **AO1**

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) $1.5 = 0.7 \times \lambda$ rearrangement and evaluation 2.1(4) m	 $\frac{1.5}{0.7}$ allow $\frac{0.7}{1.5}$ for 1 mark award full marks for correct answer without working. $\lambda = v/f$ scores 1 mark	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(iii)	A description to include: mention of oscillations/vibrations (1) EITHER transverse – (oscillations) perpendicular to direction of wave (travel) (1) OR longitudinal – (oscillations) in same direction as wave (travel) (1)	up and down OR side to side (movements) OR back and forth transverse movement up and down but longitudinal is side to side (1 mark only)	(2) AO1

Q4.

Question Number	Answer	Additional guidance	Mark
(i)	evidence of use of scale on horizontal distance axis only (1) 12 (cm) (1)	may be seen on the diagram range 11.5 to 12.5 (cm) award full marks for the correct answer without working 6 (cm) or 30(cm) scores 1 mark (evidence of use)	(2)

Question Number:	Answer	Additional guidance	Mark
(ii)	a description to include: moves up and down (1) at right angles / normal / perpendicular to (direction of) wave/travel (1)	independent marking points vertical (oscillations) not in the (direction of) wave / travel accept 'transverse wave' for 2nd MP	(2)

Q5.

Question number	Answer	Additional guidance	Marks
(i)	speed = frequency × wavelength (1)	allow v (or c) = fλ or correct equivalent, with other recognisable symbols, including w for wavelength ignore triangles	(1)

Question number	Answer	Additional guidance	Marks
(ii)	substitution (1) 0.40 × 1.2 evaluation (1) 0.48 (m/s)	apply ecf from an incorrect equation in i (or in ii) for 1 mark maximum accept 0.5 (m/s) correct answer without working gets 2 marks	(2)

Q6.

	Answer	Additional guidance	Mark
(i)	12		(1) AO1

	Answer	Additional guidance	Mark
(ii)	$\frac{42}{12}$ (1) 3.5 (cm) (1)	 ecf from2ai allow 0.035 for 1 mark award full marks for the correct answer without working	(2) AO1

	Answer	Additional guidance	Mark
(iii)	A description to include: **either** time a crest/ripple/wavefront (1) (moving) between **P** and **Q** (1) use (wave speed =) $\frac{\text{distance}}{\text{time}}$ (1) **or** count number of crests /ripples /wavefronts passing (eg P) (1) in a given time (to find f) (1) use (v =) f λ (1)	 allow 'how long it takes' allow 'wave' for crest allow – over the 42 cm over a (set) distance allow waves if no other mark scored measure frequency for 1 mark	(3) AO1

Q7.

Q8.

Question Number	Answer	Additional guidance	Mark
(i)	recall speed = $\frac{\text{distance}}{\text{time}}$	accept any correct rearrangement or use of s, d and t may use v for speed and x for distance ignore use of triangles	**(1)** AO 1 1

Question Number	Answer	Additional guidance	Mark
(ii)	substitution (1) (speed) = $\frac{220}{0.7(0)}$ evaluation (1) 310 (m/s)	allow ecf from part (i) for this mark only allow any numbers that round to 310 e.g. 314 award full marks for the correct answer without working	**(2)** AO 2 1

Q9.

Question number	Answer	Additional guidance	Mark
(i)	at least one arrow in the direction QS (1) two arrows in opposite directions (1)	allow arrows parallel to QS independent mark scores 2 marks two arrows in opposite directions but perpendicular to QS scores 1 mark maximum	(2)

Question number	Answer	Additional guidance	Mark
(ii)	converts 7 km/s to 7000 m/s (1) substitution (1) $\frac{7(\times 10^3)}{12}$ evaluation (1) 580 (m)	7000 seen (1) allow numbers that round down to 580 such as 583.33.... 5.8 to any incorrect power of ten scores 2 marks award full marks for the correct answer without working	(3)

Q10.

	Answer	Acceptable answers	Mark
(a)(i)	☒ D ROS		(1)

(a)(ii)	☒C changes changes		**(1)**
(b)(i)	An explanation linking **three** of the following: • the (objective) lens (1) • collects light (from Jupiter) (1) • focuses/converges/refracts the light (1) • (objective) produces a real image (1) • (at the) focal point (1) • (Image is viewed by the) eyepiece lens (1) • (Eyepiece) produces a virtual image (1)	Marks can be awarded for a labelled diagram	**(3)**
(b)(ii)	An explanation linking the following: (the geocentric model) • has the Earth at the centre of the Universe/solar system / everything orbits the Earth (1)(but Galileo observed) • moons of Jupiter (1) • (so) not all things orbit the Earth/ geocentric model is wrong (1)	**Accept alternative argument:**Galileo observed moons/stars/objects Orbiting/going round Jupiter (this) suggested that not everything orbits the Earth	**(3)**
(c)	2100/500 = 4.2 (1) 4.2 × 150 = 630 (million km) (1) Accept ratios as speed is constant 150/500 = distance to Jupiter/2100 OR Distance to Jupiter = (150/500) × 2100 Either for 1 mark	Power of 10 error maximum of 1 mark (speed of light) about 150 000 000 ÷ 500 = 300 000 (km/s) (1) (distance to Jupiter)= 300 000 × 2 100 = 630 000 000 km (1) / = 630 (million km) An answer with no calculation of 630 (million km) gains 2 marks If an answer	**(2)**

		of 630 million/ 630 000 000 is given with correct working award both marks	

Q11.

Question Number	Answer	Acceptable answers	Mark
(i)	**X** amplitude (1) **Y** wavelength (1)		**(2)**

Question Number	Answer	Acceptable answers	Mark
(ii)	**A** (1)		**(1)**